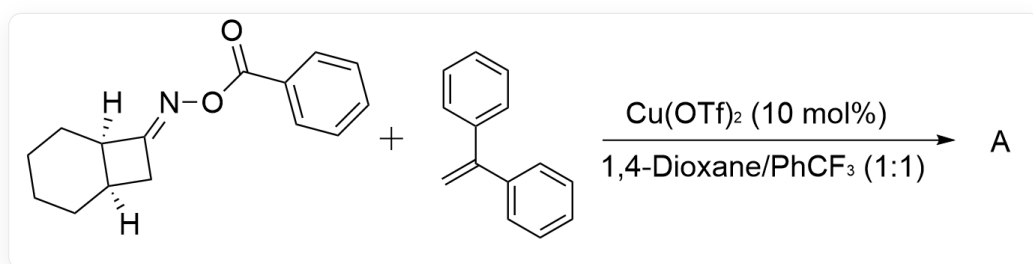


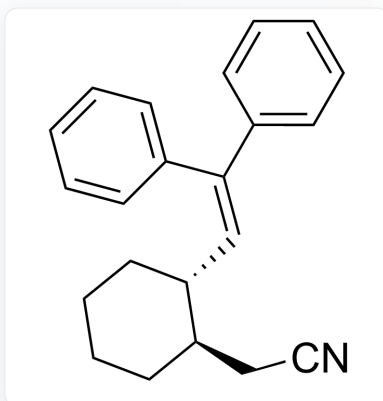
Question



[H][C@@]1(CCCC2)[C@@]2([H])C/C1=NOC(C3=CC=CC=C3)=O.C=C(C1=CC=CC=C1)C2=CC=CC=C2> $\text{Cu}(\text{OTf})_2$ (10 mol%),
1,4 - Dioxane/ PhCF_3 (1 : 1)>[A]

Please select the most suitable product A and predict whether the Z/E configuration of the carbon-nitrogen double bond will affect the reaction selectivity.

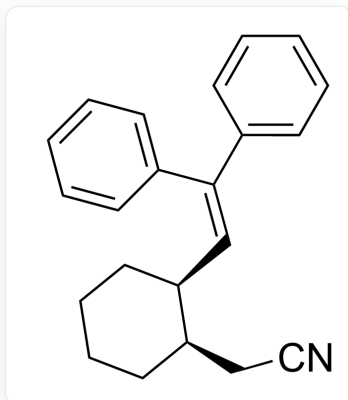
A.



N#CC[C@@H]1[C@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

No

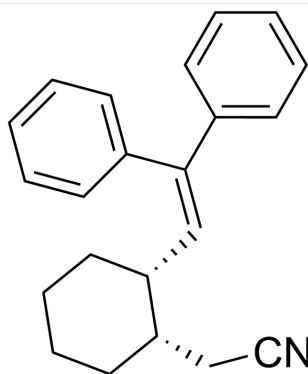
B.



N#CC[C@@H]1[C@@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

No

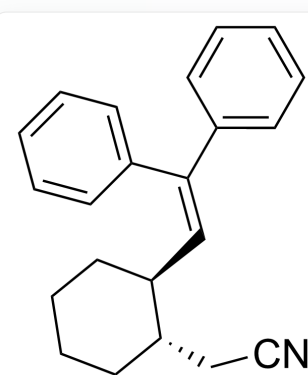
C.



N#CC[C@H]1[C@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

Yes

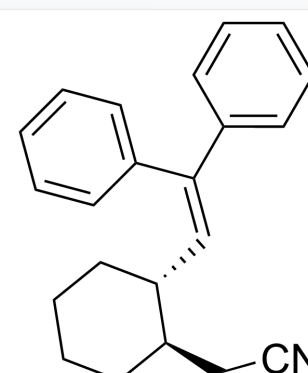
D.



N#CC[C@H]1[C@@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

Yes

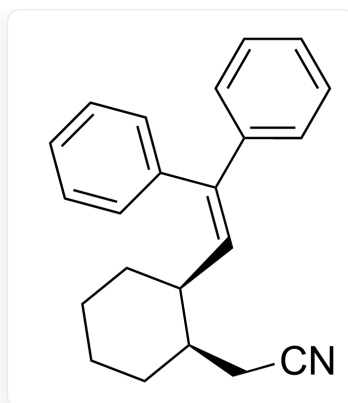
E.



N#CC[C@@H]1[C@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

Yes

F.



N#CC[C@@H]1[C@@H](CCCC1)/C=C(C2=CC=CC=C2)/C3=CC=CC=C3

Yes

Answer

Correct Answer: A

Detailed Explanation

This reaction is a free radical reaction

CHECKPOINT

1 PTS

This reaction is a free radical reaction

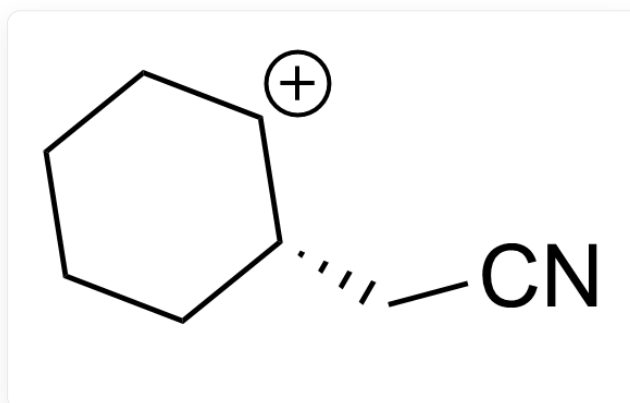
Cu in the reaction system is a catalyst

CHECKPOINT

1 PTS

Cu in the reaction system is a catalyst

Cu(I) first donates a single electron, causing the weaker N-O bond to cleave, forming a free radical intermediate. The cleavage of the nitrogen – oxygen single bond is not the source of stereoselectivity, so the *Z/E* configuration of Cu(II) oxidizes this radical to form a carbocation



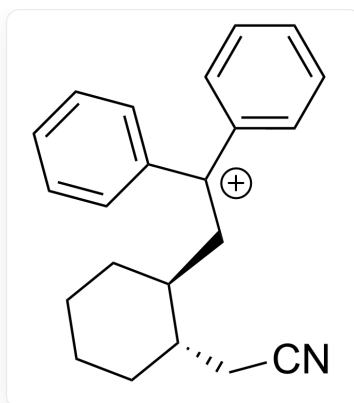
N#CC[C@H]1[CH+]CCCC1

CHECKPOINT

1 PTS

Then Cu(II) oxidizes this radical to form a carbocation

Since the steric hindrance above this carbocation is greater than that below, it is selectively captured by stilbene from below



N#CC[C@H]1[C@@H](CCCC1)C[CH+](C2=CC=CC=C2)C3=CC=CC=C3

CHECKPOINT

1 PTS

Since the steric hindrance above this carbocation is greater than that below, it is selectively captured by stilbene from below

Finally, deprotonation forms product A